Review on data-driven computational mechanics

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22 January 2021

Concept of data-driven computational mechanics

- Michael Ortiz (Caltech)

  Frank and Ora Lee Marble Professor of Aeronautics and Mechanical Engineering

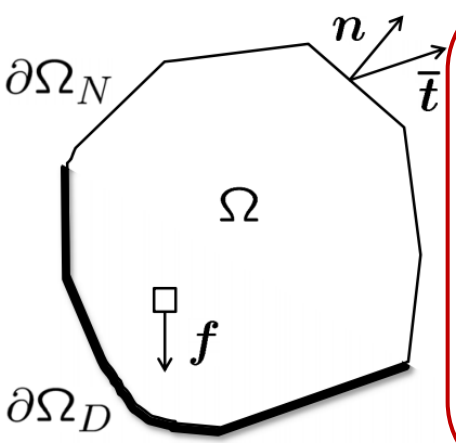


- Publications on data-driven mechanics
  - Karapiperis, K. et al. (2021) *Data-Driven Multiscale Modeling in Mechanics*, JMPS
  - Eggersmann, R. et al. (2021) *Model-free data-driven computational mechanics enhanced by tensor voting*, CMAME
  - Carrara, P. et al. (2020) *Data-driven fracture mechanics*
  - Conti, S. et al. (2020) *Data-Driven Finite Elasticity*
  - Stainier, L. et al. (2019) *Model-free data-driven methods in mechanics: material data identification and solvers*, Comp. Mech.
  - Eggersmann, R. et al. (2019), *Model-Free Data-Driven inelasticity*
  - Kirchdoerfer, T.,Ortiz, M. (2018) *Data-Driven Computing in Dynamics*
  - Kirchdoerfer, T.;Ortiz, M. (2017) *Data Driven Computing with Noisy Material Data Sets*
  - Kirchdoerfer, T.;Ortiz, M. (2016) *Data-driven computational mechanics*, CMAME

Concept of data-driven computational mechanics

- Three governing equations

Universal laws (material-independent)



i) Kinematics + Dirichlet:

$$\epsilon(u) = 1/2(\nabla u + \nabla u^T)$$
$$u = \bar{u}, \quad \text{on } \partial\Omega_D$$

ii) Equilibrium + Neumann:

$$\text{div}\,\sigma + f = 0$$
$$\sigma n = \bar{t}, \quad \text{on } \partial\Omega_N$$

iii) Material law: $\sigma(x) = \sigma\big(\epsilon(x)\big)$

Unknown

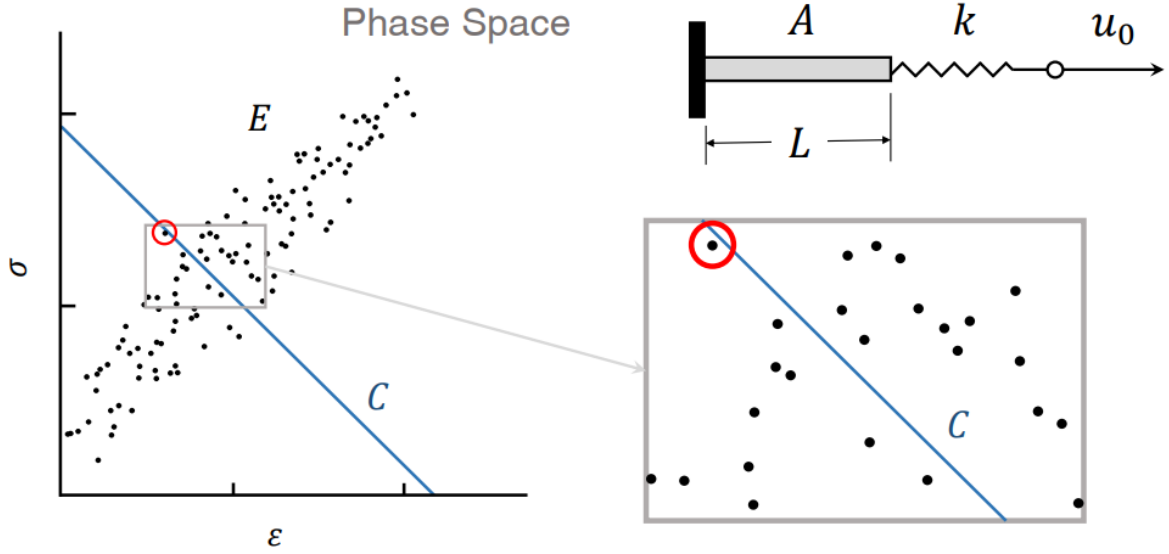
Conventional approach:

Physical or phenomenal models for different material

Data-driven (model-free) approach:

Universal framework purely based on data

- 1D bar example



- **Stress-strain phase space : $(\sigma, \varepsilon) \in R^2$**
- **Constraint set (kinematics + equilibrium)**

$$C = \{(\sigma, \varepsilon): \sigma A = k(u_0 - \varepsilon L)\}$$

- **Material data set**

$$E = \{(\sigma_e, \varepsilon_e)\}$$

- **Solution set: $C \cap E$**

Search for closest data point in material set (distance-minimizing algorithm)

➢ Defined norm for distance measuring

$$F_e(\varepsilon_e, \sigma_e) = \min_{(\varepsilon'_e, \sigma'_e) \in E_e} \Big(W_e(\varepsilon_e - \varepsilon'_e) + W_e^*(\sigma_e - \sigma'_e) \Big),$$

$$W_e(\varepsilon_e) = \frac{1}{2} C_e \varepsilon_e^2 \quad \text{and} \quad W_e^*(\sigma_e) = \frac{1}{2} \frac{\sigma_e^2}{C_e}$$

➢ Optimization problem

$$\text{Minimize:} \quad \sum_{e=1}^{m} w_e F_e(\varepsilon_e, \sigma_e),$$

$$\text{subject to:} \quad \varepsilon_e = \sum_{i=1}^{n} B_{ei} u_i \quad \text{and} \quad \sum_{e=1}^{m} w_e B_{ei} \sigma_e = f_i,$$

➢ Lagrange solution

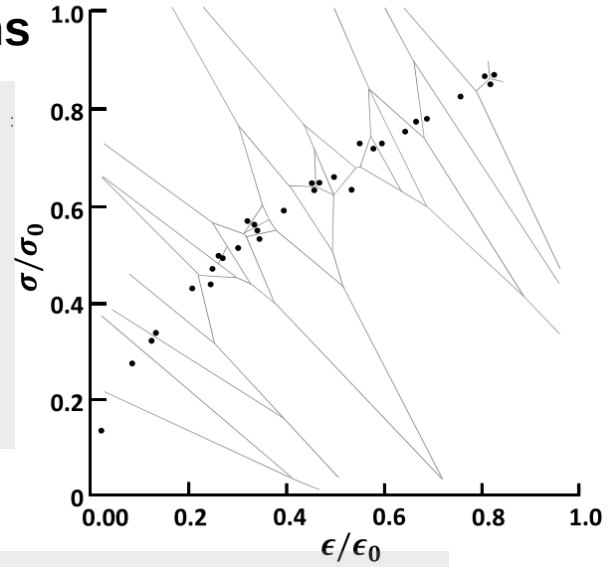
$$\delta \left(\sum_{e=1}^{m} w_e F_e\Big(\sum_{i=1}^{n} B_{ei} u_i, \sigma_e \Big) - \sum_{i=1}^{N} \Big(\sum_{e=1}^{m} w_e B_{ei} \sigma_e - f_i \Big) \eta_i \right) = 0.$$

➢ System of linear equations

$$\delta u_i \Rightarrow \sum_{e=1}^{m} w_e C_e \Big(\sum_{j=1}^{n} B_{ej} u_j - \varepsilon_e^* \Big) B_{ei}$$

$$\delta \sigma_e \Rightarrow \frac{1}{C_e} (\sigma_e - \sigma_e^*) = \sum_{i=1}^{n} B_{ei} \eta_i,$$

$$\delta \eta_i \Rightarrow \sum_{e=1}^{m} w_e B_{ei} \sigma_e = f_i,$$



➢ Algorithm

Given: material data set *E*, B matrix

i) Chose $\left(\sigma^{*(0)}, \varepsilon^{*(0)}\right)$ randomly from *E*

ii) Solve linear equations for $\left(\sigma^{(k)}, \varepsilon^{(k)}\right)$ Global step

iii) Chose $\left(\sigma^{*(k)}, \varepsilon^{*(k)}\right)$ closet to $\left(\sigma^{(k)}, \varepsilon^{(k)}\right)$ form *E*

$$\|(\varepsilon_e, \sigma_e)\|_e = \left(W_e(\varepsilon_e) + W_e^*(\sigma_e) \right)^{1/2}$$ Local step

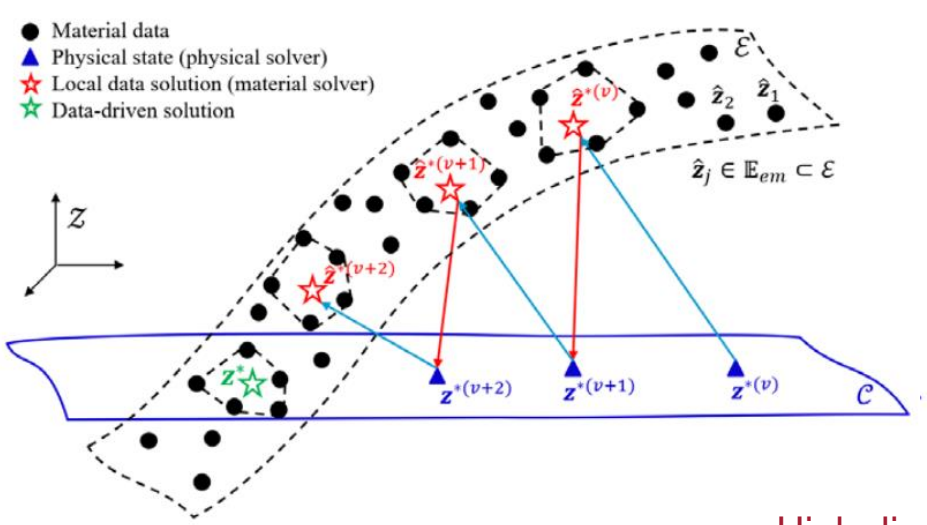
iv) Convergence test

If: $\left(\sigma^{*(k+1)}, \varepsilon^{*(k+1)}\right) = \left(\sigma^{*(k)}, \varepsilon^{*(k)}\right)$, converge

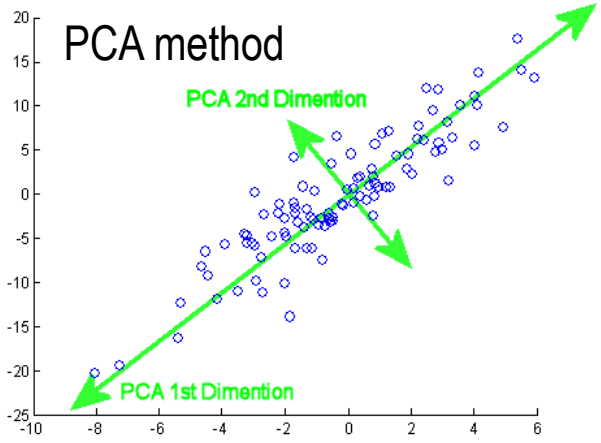
Else: $k \leftarrow k + 1$, **goto** ii)

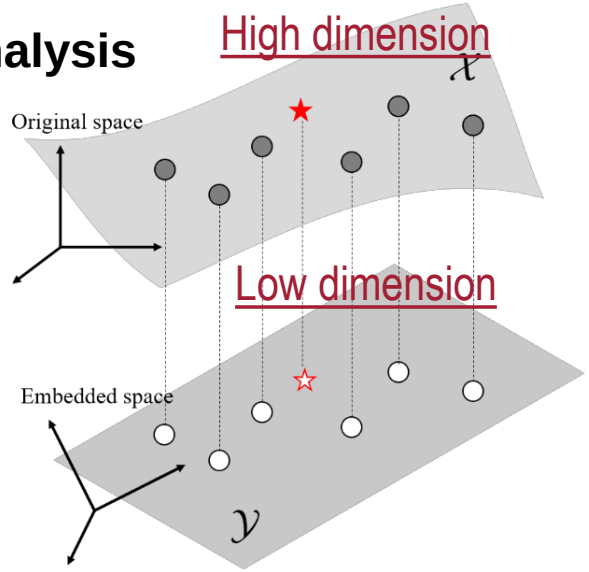
Manifold learning (Nonlinear dimensionality reduction) dealing with noisy data

- **High-dimensional distance-minimizing**



- **Dimension reduction analysis**



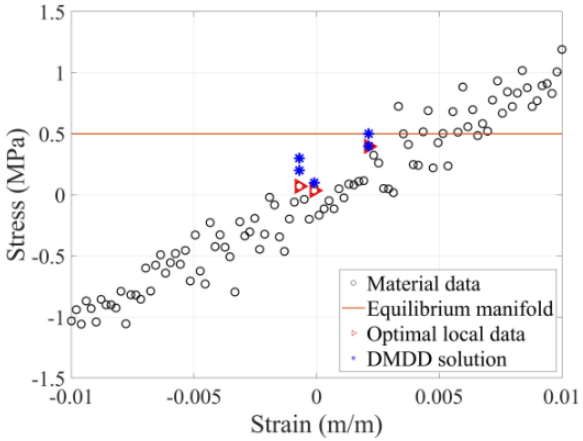


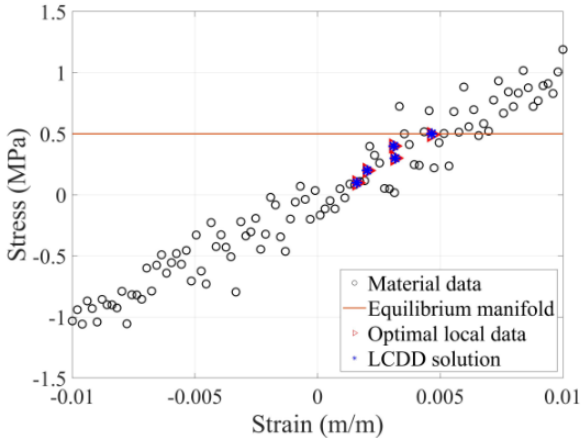
- **Manifold learning (filter out noise)**

  Map high-dimensional data to low-dimensional

- **Locally convex reconstruction**

  Projection to convex hull of k nearest neighbor points





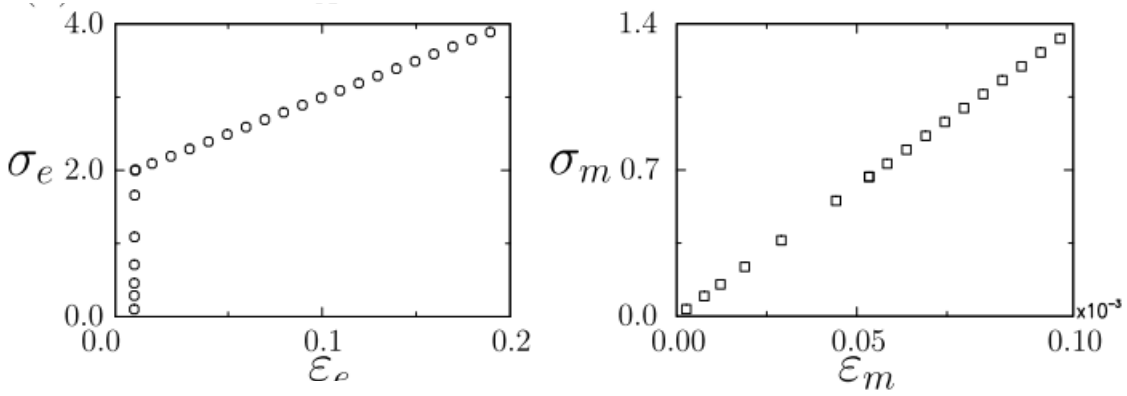
Ibañez, R., et al., "A manifold learning approach to data-driven computational materials and processes," AIP Conf. Proc. 1896, 2017

He, Q. et al., A physics-constrained data-driven approach based on locally convex reconstruction for noisy database, ArXiv, 2019.

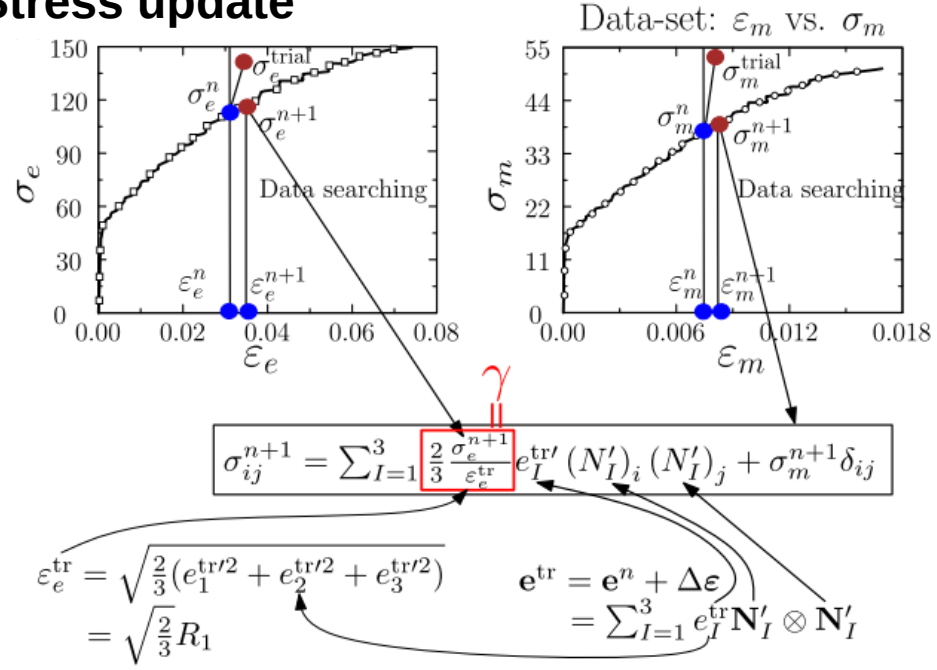
He, X., et al. "Physics-constrained local convexity data-driven modeling of anisotropic nonlinear elastic solids," Data-Centric Eng., 2020

Mapping 1D data to 3D (MAP123 data-driven model)

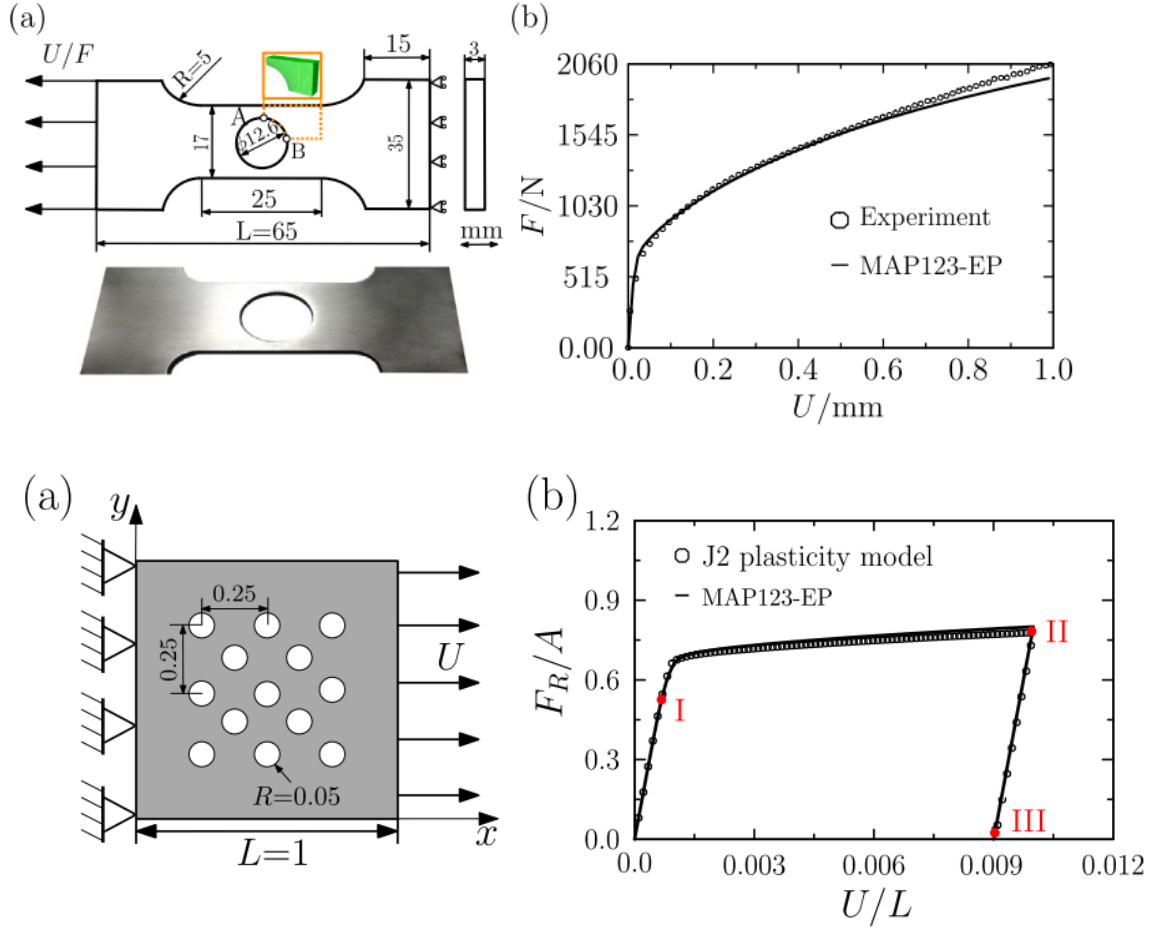
- **Material data set (J2 plasticity)**



- **Stress update**



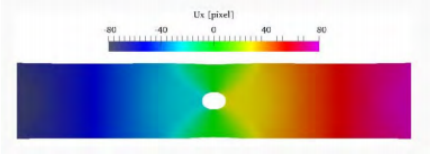
- **Results**



Tang, S., et al. "MAP123-EP: A mechanistic-based data-driven approach for numerical elastoplastic analysis," *Comput. Methods Appl. Mech. Eng., 2020*

Material data identification full-field experimental data

➢ **Full-field data from central-hole tension)**



- Full-field measurements (DIC), M loading cases:

$$D_{\text{exp}} = \{(\boldsymbol{u}^\alpha, \boldsymbol{f}^\alpha),\ \alpha = 1, \ldots, M\}.$$

- Stresses σ_e^α cannot be measured directly!
- Material-data set:

$$D = \cup_{\alpha=1}^{M} \{\epsilon_e = B_e u^\alpha, \textstyle\sum_{e=1}^{m} w_e B_e^T \sigma_e = f^\alpha\}$$

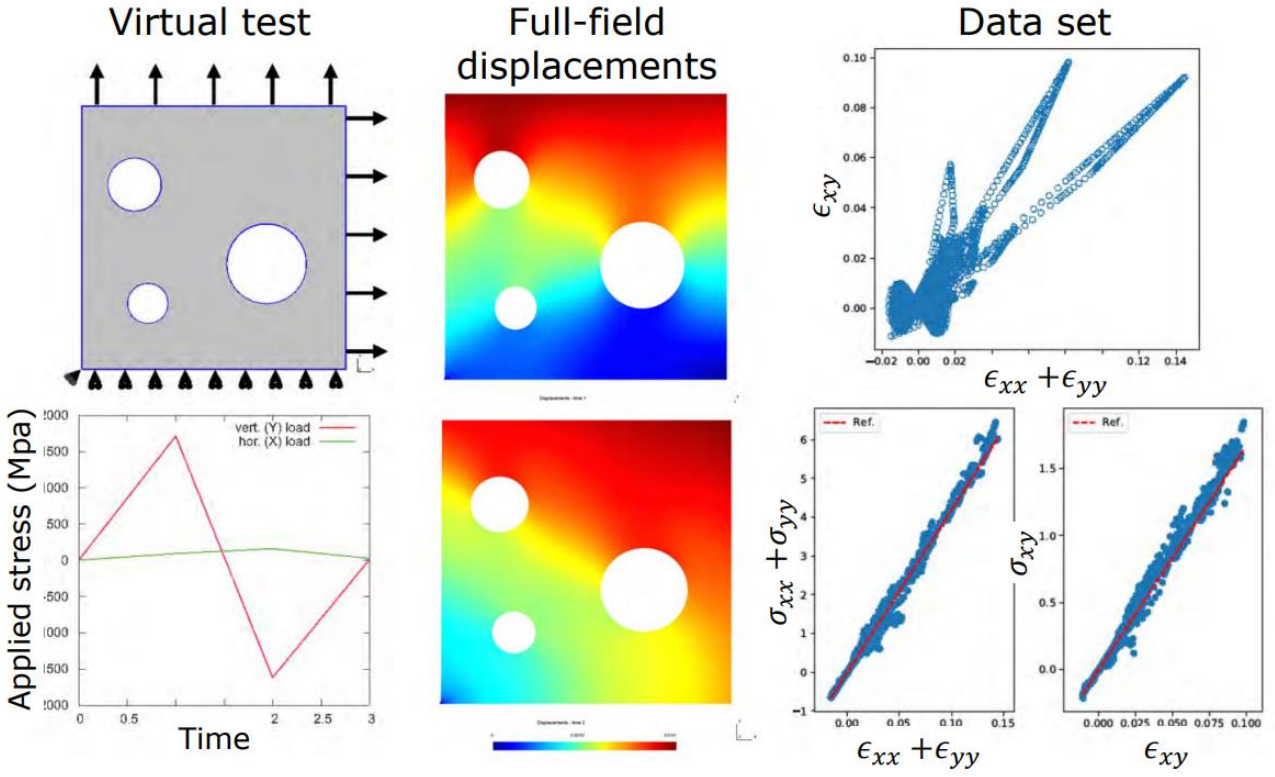
- DD self-consistent material-data identification:

$$\text{Given } D_{\text{exp}}: \quad \min_{(\boldsymbol{\epsilon}^*, \boldsymbol{\sigma}^*) \in D} \left(\min_{(\boldsymbol{\epsilon}, \boldsymbol{\sigma}) \in E} |(\boldsymbol{\epsilon} - \boldsymbol{\epsilon}^*, \boldsymbol{\sigma} - \boldsymbol{\sigma}^*)|^2 \right)$$

J. Rethore, HAL Id: hal-01454432, Feb. 2017.
J. Rethore and A. Leygue, HAL Id: hal-01452494, Feb. 2017.

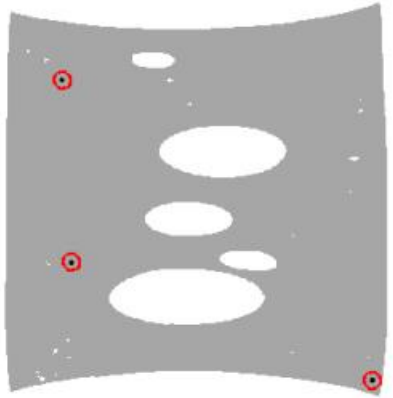
Michael Ortiz
GALCIT 2020

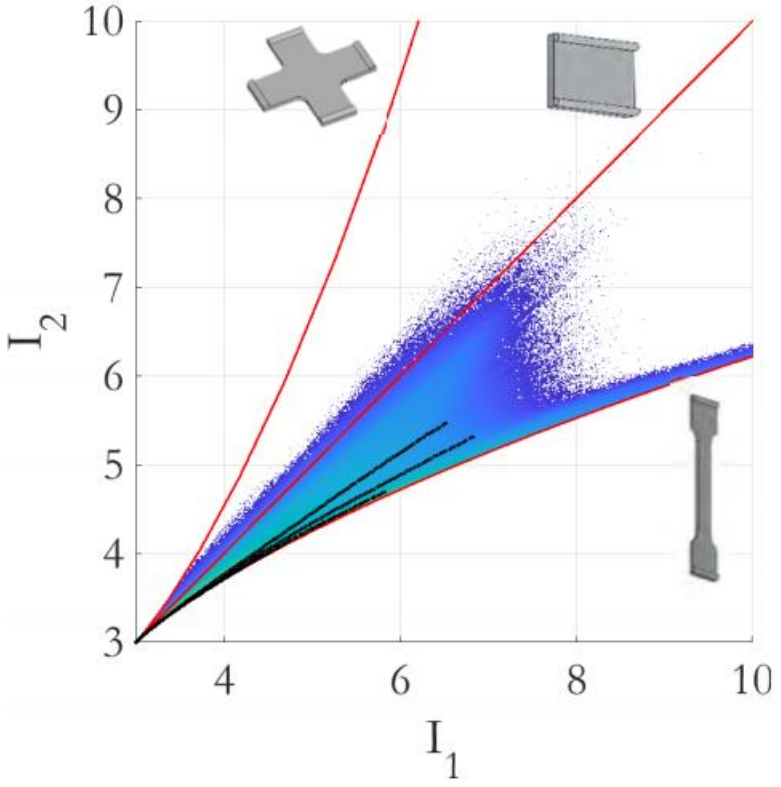
➢ **Calibration results**



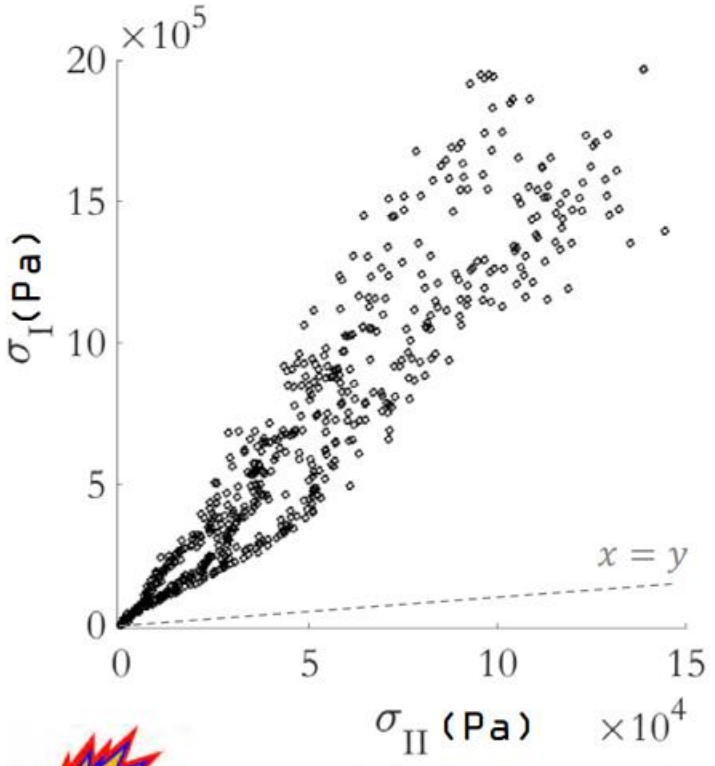
L. Stainier, A. Leygue and M. Ortiz, *Comp. Mech.*, **64** (2019)(2) .

Instructions on test: wider stress-strain space with perforated sample





Strain data





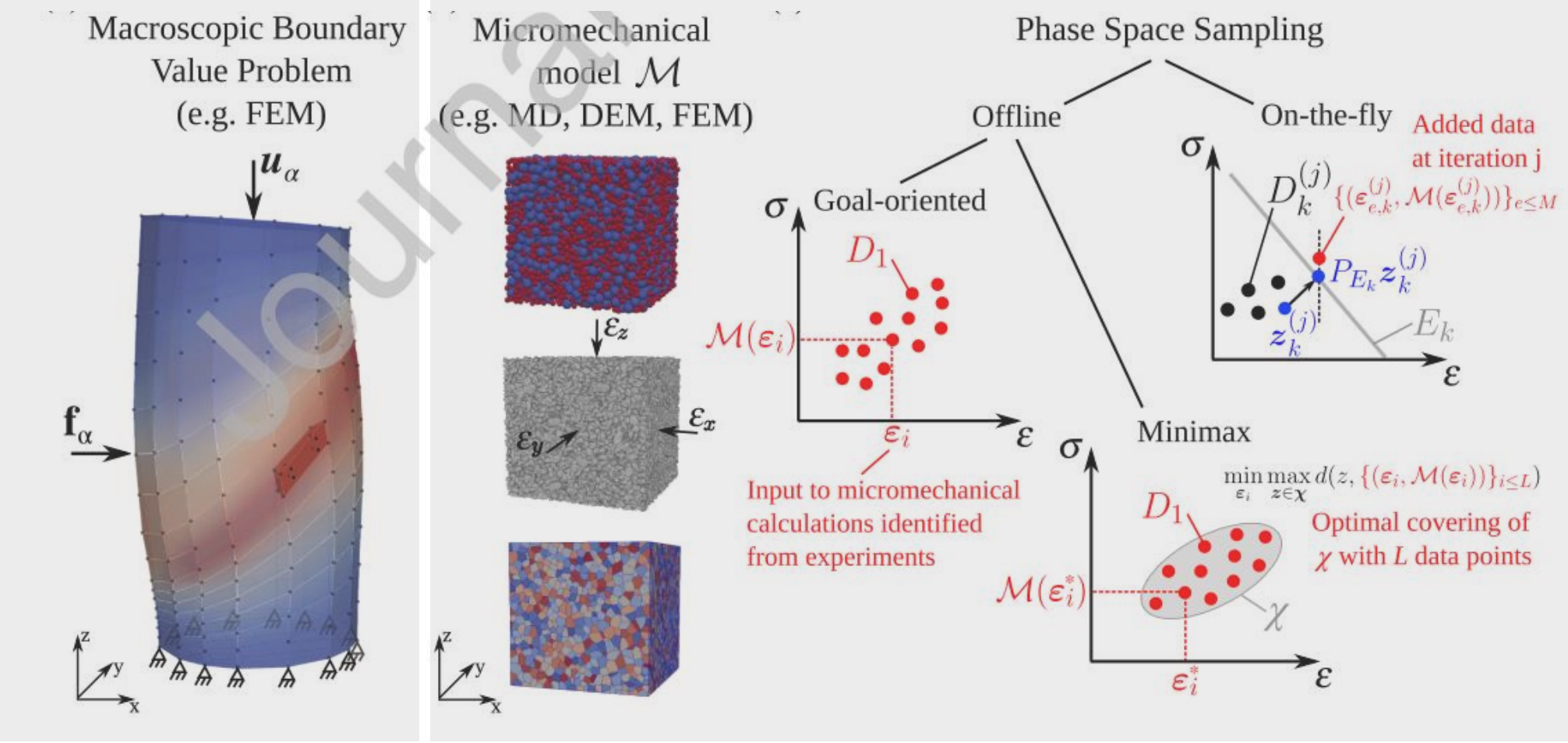
Stress data

(M. Dalémat, M. Coret, A. Leygue and E. Verron, 2019)

Michael Ortiz
GALCIT 2020

Data-Driven multiscale modeling

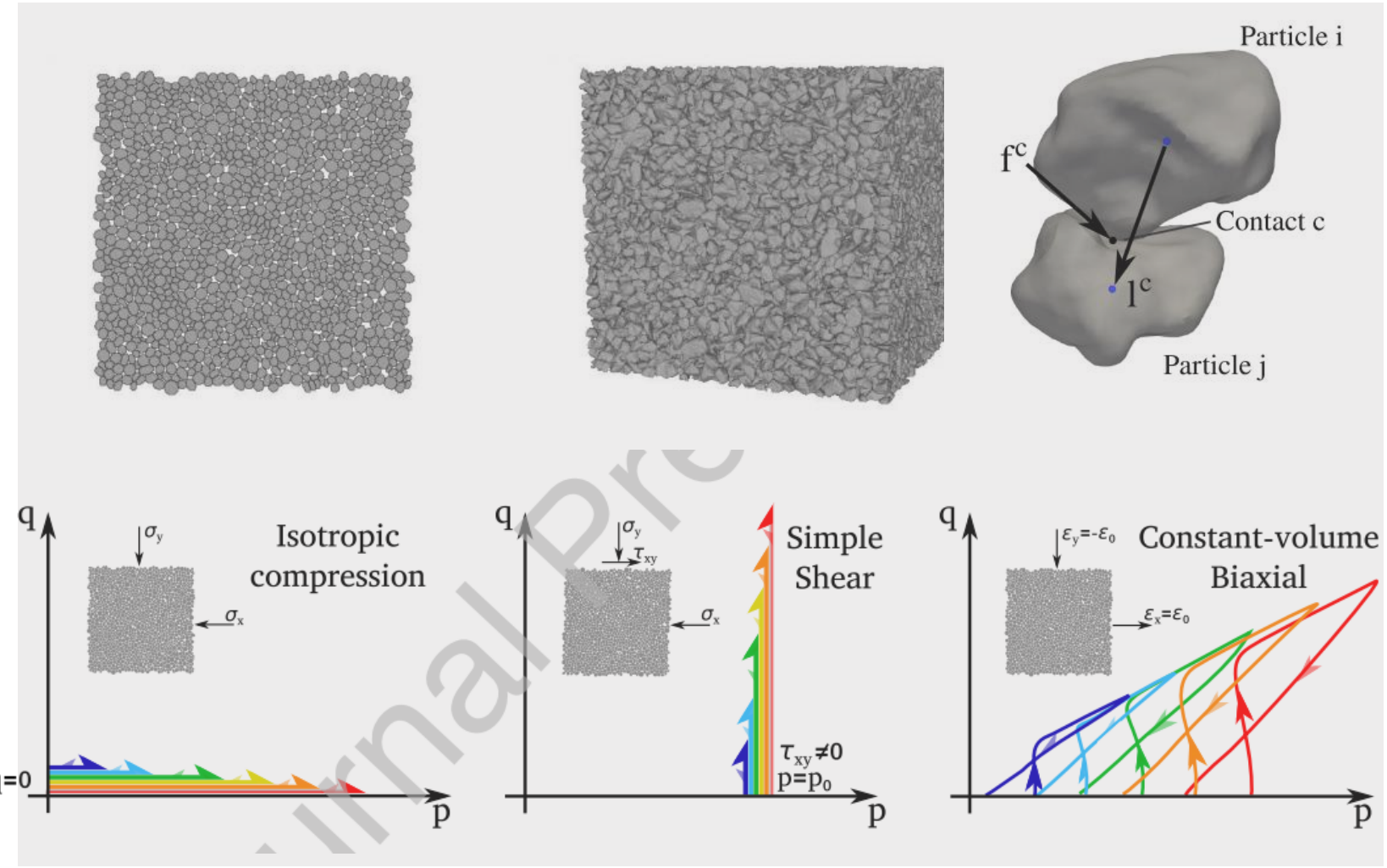
- **Data-driven macro model**
- **Three strategies generating material data with micro model**



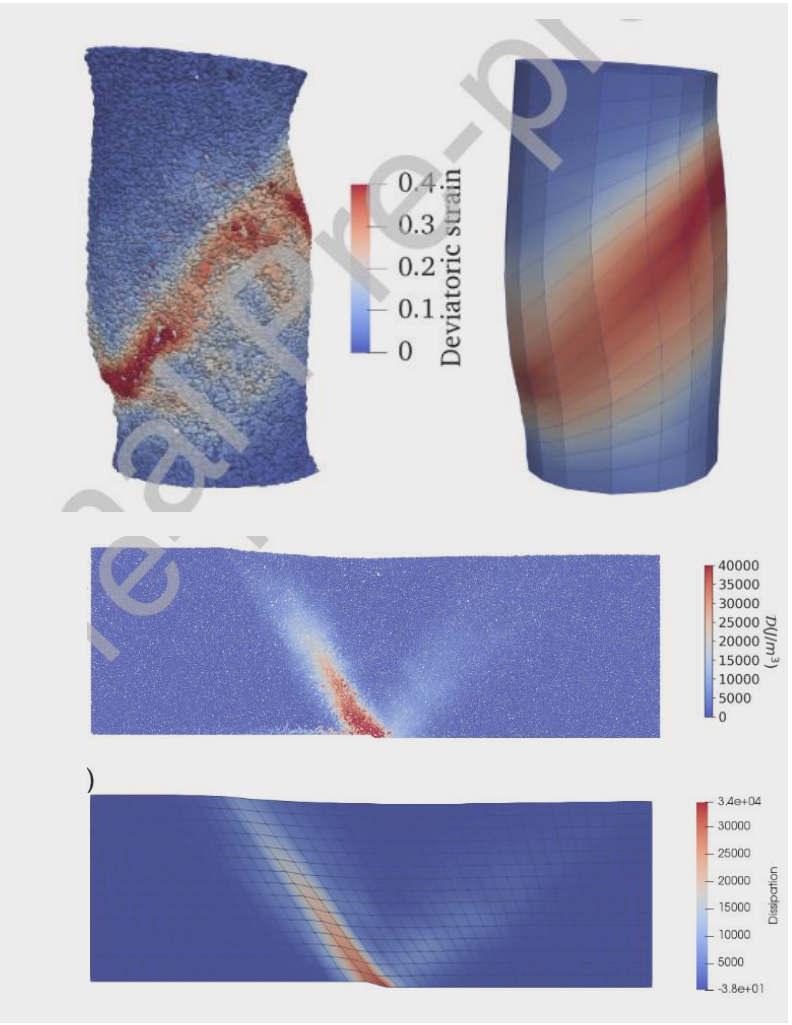
Karapiperis, K., et al, "Data-Driven multiscale modeling in mechanics," *J. Mech. Phys. Solids, 2020*

Data-Driven multiscale modeling

- **Level-Set Discrete Element Method (LS-DEM) simulations**



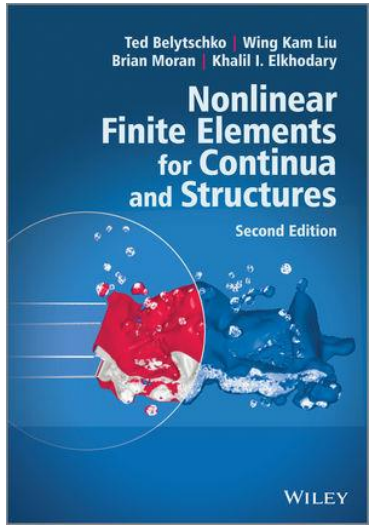
- **Results**



Hierarchical Deep-learning Neural Networks (HiDeNN)

- Wing Kam Liu (Northwestern Uni.)

Walter P. Murphy Professor of Mechanical Engineering & Civil and Environmental Engineering

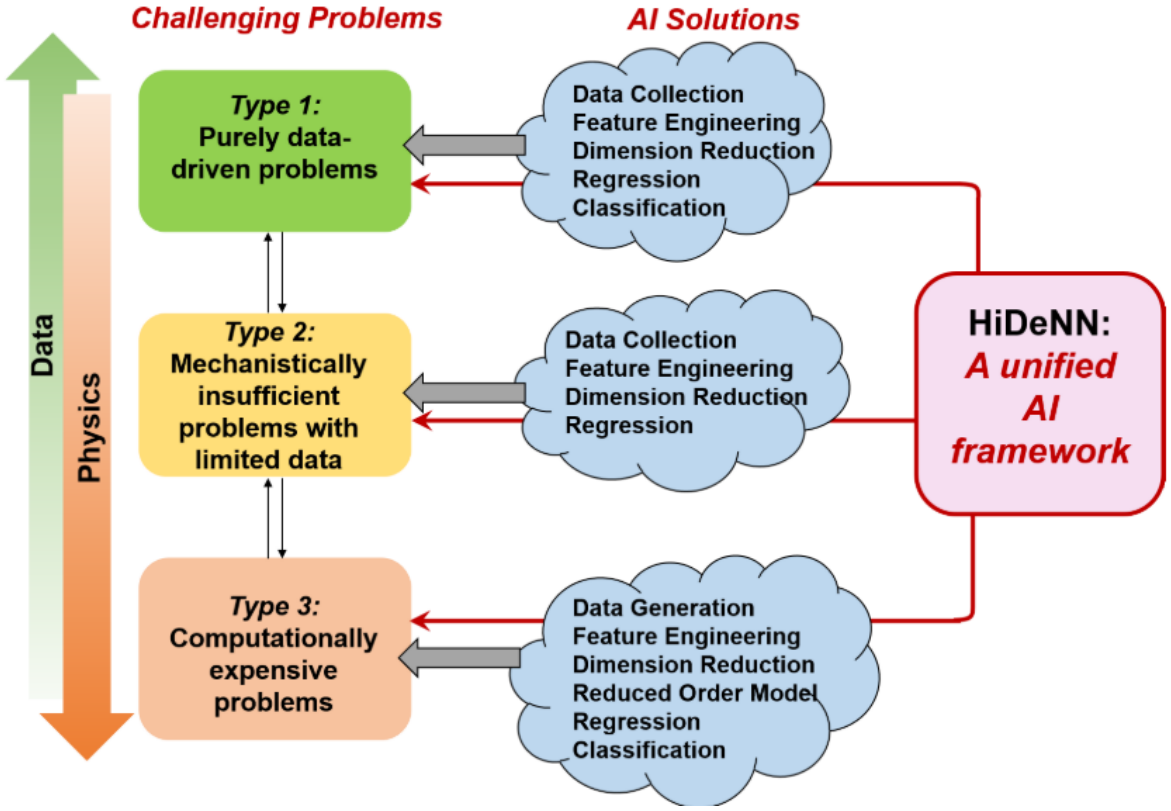


- Publications on data-driven mechanics

Zhang, L., et al. "Hierarchical deep-learning neural networks: finite elements and beyond," *Comput. Mech.*, 2020.

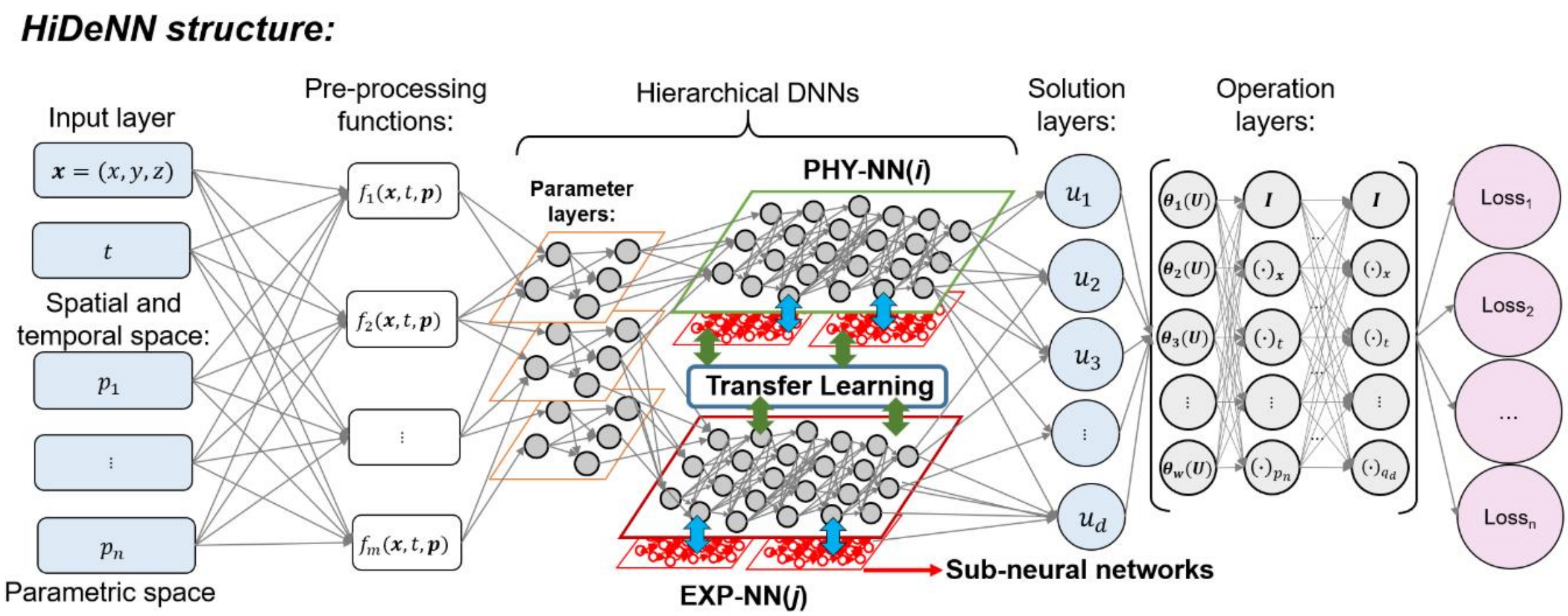
Saha, S., et al. "Hierarchical Deep Learning Neural Network (HiDeNN): An artificial intelligence (AI) framework for computational science and engineering," *Comput. Methods Appl. Mech. Eng.* 373(October), 2021.

- Three types of challenging problems

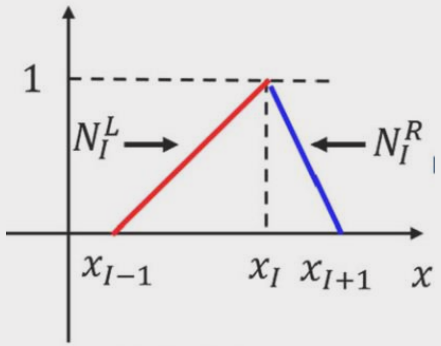


- **Type 1: purely data-driven (sufficient data)**
- **Type 2: mechanistically insufficient with limited data**
- **Type 3: computationally expensive**

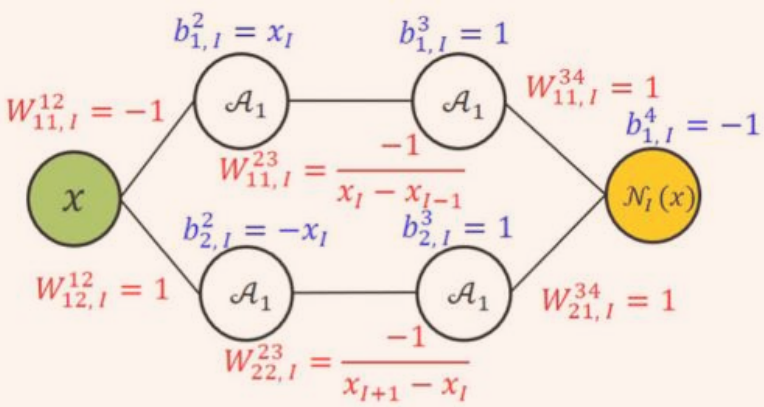
Hierarchical deep-learning neural networks (HiDeNN)



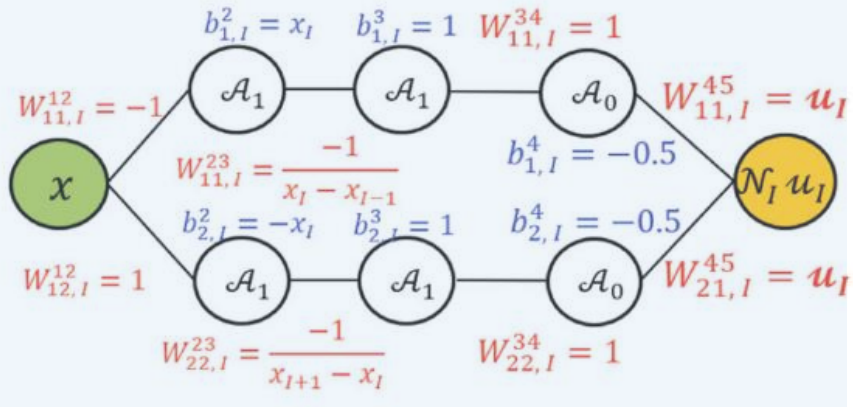
Construction of HiDeNN



Global Shape function at x_I



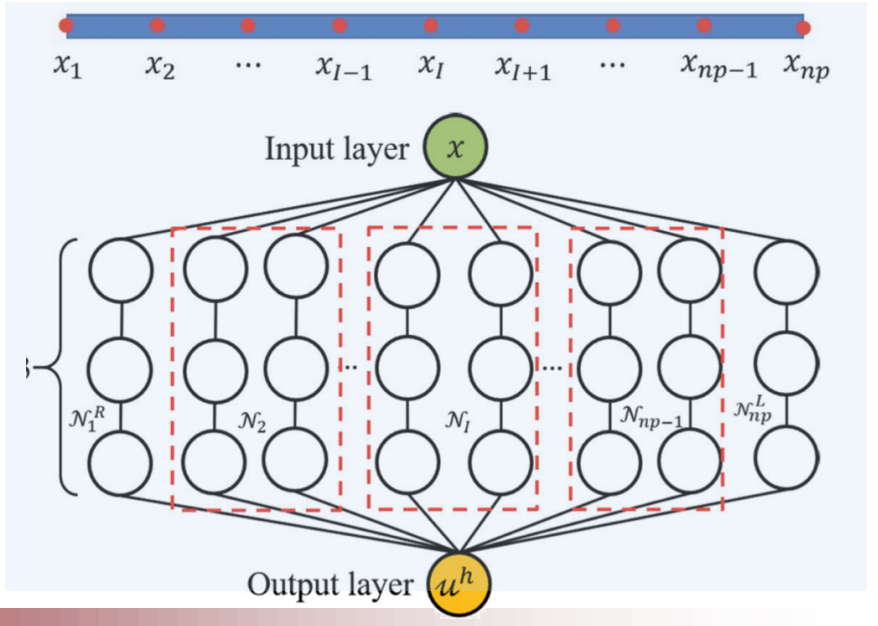
(a) DNN-based 1D shape function



(b) DNN-based 1D interpolation function

$$N_I(x) = \begin{cases} \dfrac{x - x_{I-1}}{x_I - x_{I-1}}, & x_{I-1} \le x \le x_I, \\ \dfrac{x_{I+1} - x}{x_{I+1} - x_I}, & x_I \le x \le x_{I+1}, \\ 0, & elsewhere, \end{cases}$$

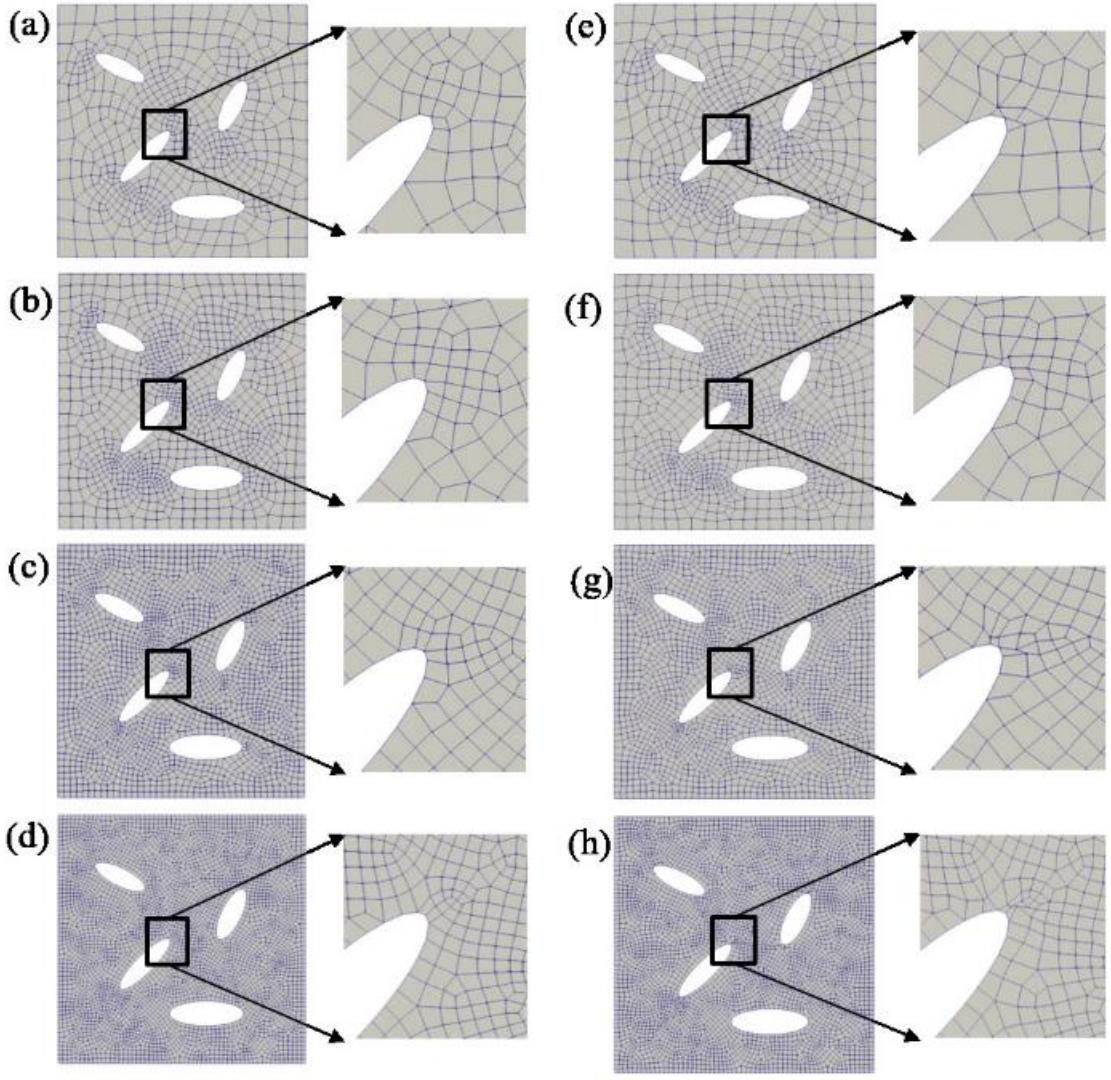
$$\begin{aligned} &\mathcal{N}_I(x;\, \boldsymbol{W}, \boldsymbol{b}, \boldsymbol{\mathcal{A}}) \\ &= W^{34}_{11,I}\mathcal{A}_1\left(W^{23}_{11,I}\mathcal{A}_1\left(W^{12}_{11,I}x + b^2_{1,I}\right) + b^3_{1,I}\right) \\ &\quad + W^{34}_{21,I}\mathcal{A}_1\left(W^{23}_{22,I}\mathcal{A}_1\left(W^{12}_{12,I}x + b^2_{2,I}\right) + b^3_{2,I}\right) + b^4_{1,I} \\ &= \mathcal{A}_1\left(\frac{-1}{x_I - x_{I-1}}\mathcal{A}_1\left(-x + x_I\right) + 1\right) \\ &\quad + \mathcal{A}_1\left(\frac{-1}{x_{I+1} - x_I}\mathcal{A}_1\left(x - x_I\right) + 1\right) - 1, \end{aligned} \tag{2}$$

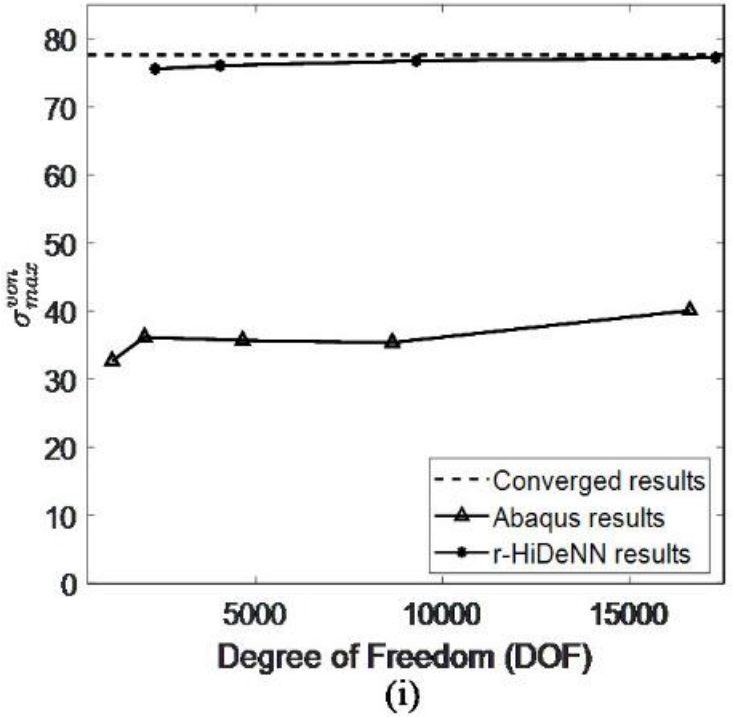


- **Replaced with NN-based shape function**
- **Parameterized with node position – adaptive mesh**

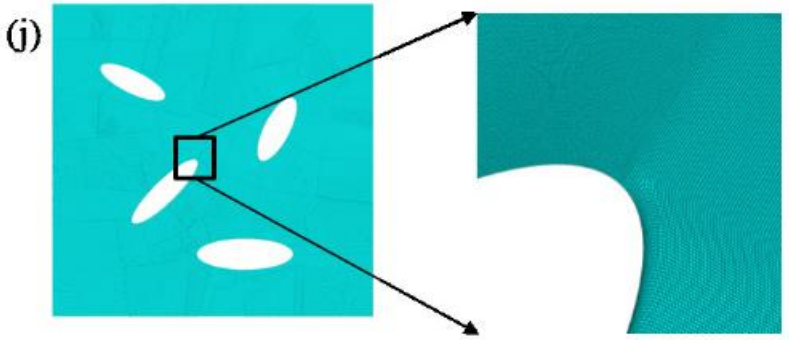
Improved accuracy with adaptive mesh

- **FEM results**
- **HiDeNN results**
- **Accuracy evaluation**



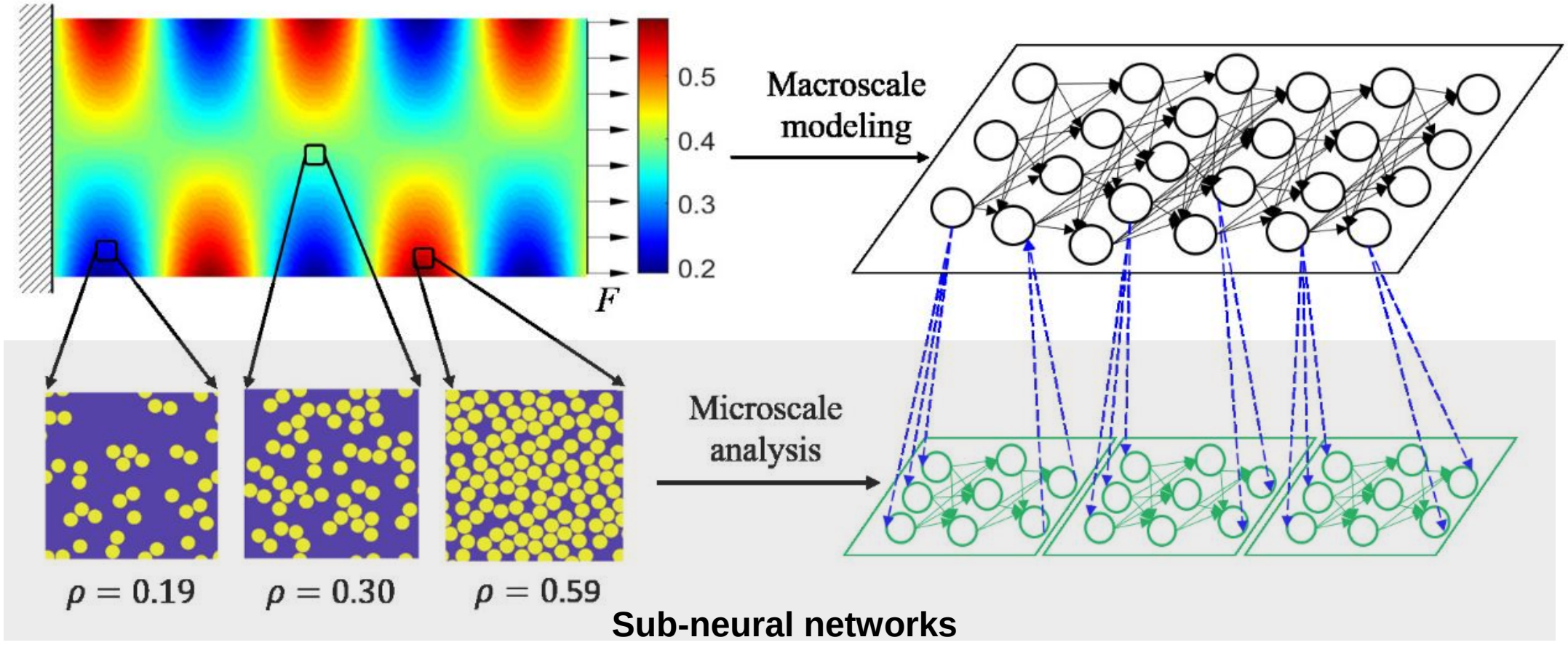


(i)



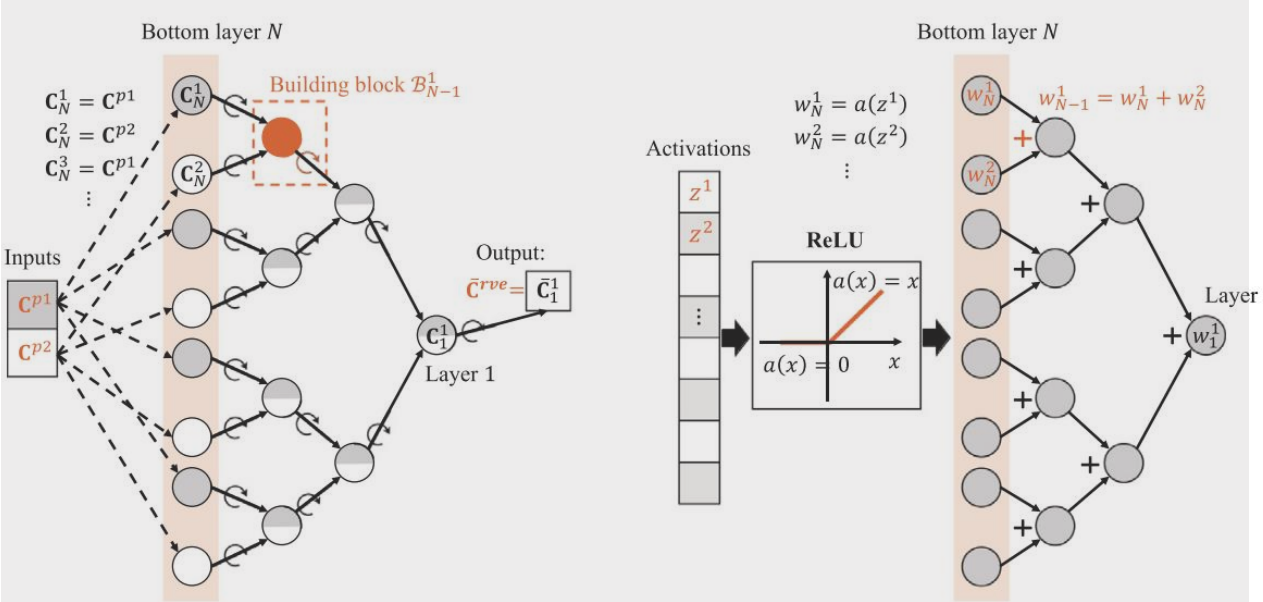
HiDeNN multi-scale modeling

- **HiDeNN framework of two-scale analysis of fiber-enforced composite materials**



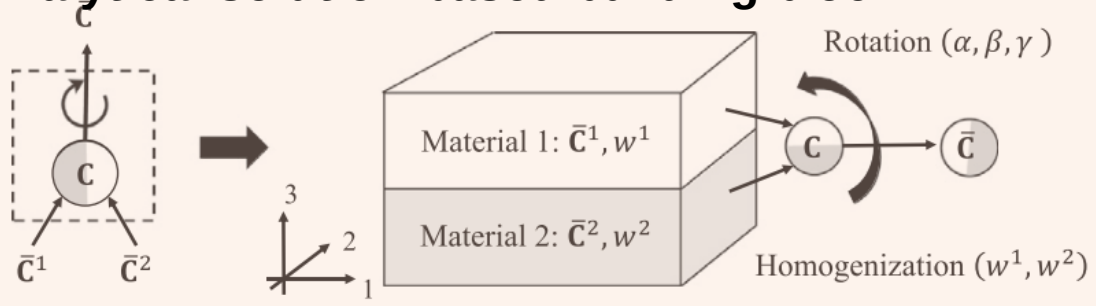
3D architectures of deep material network in data-driven multiscale mechanics

➢ Architecture of material network

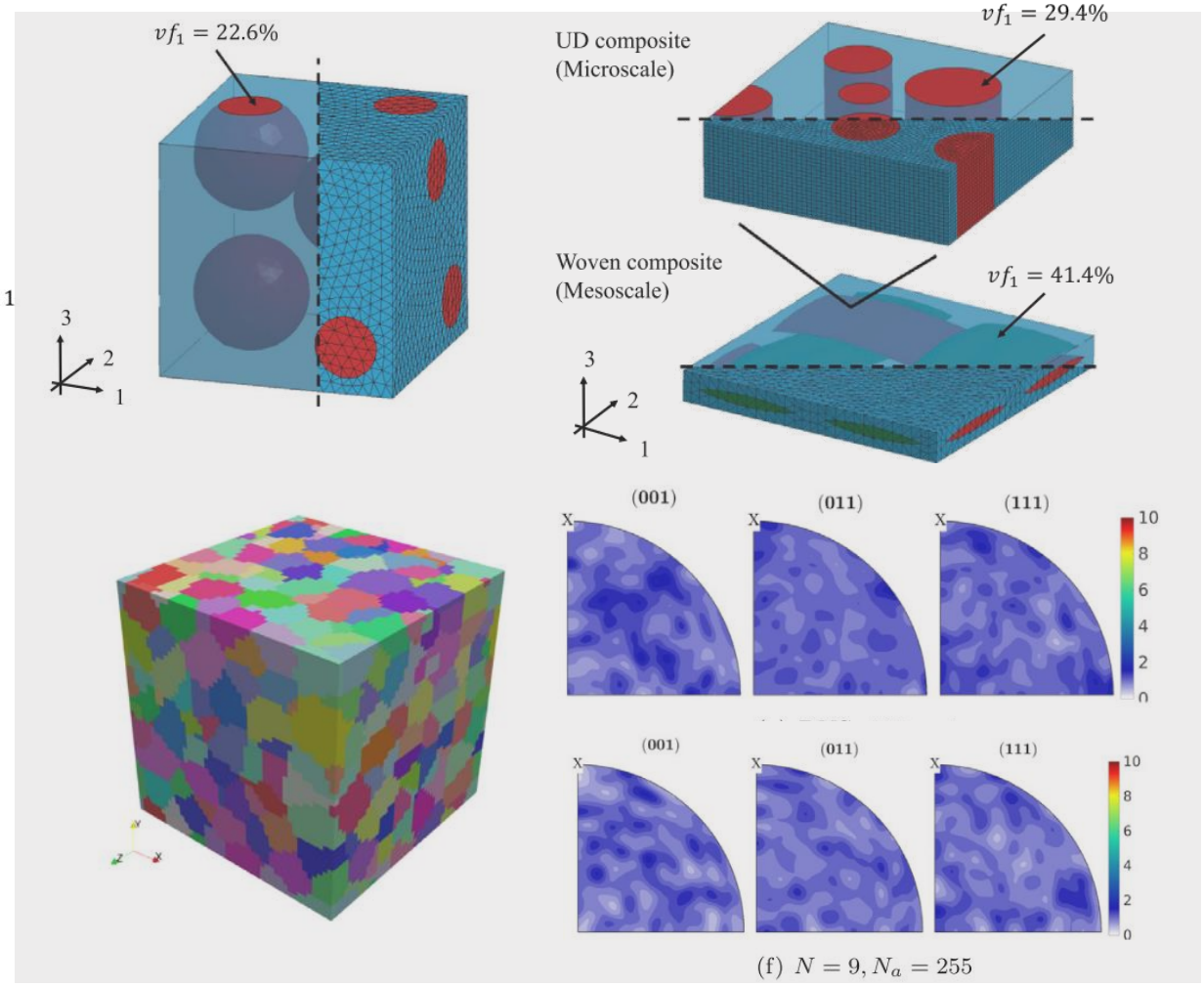


Data flow of stiffness matrix **Data flow of weights**

➢ Analytical solution-based building block

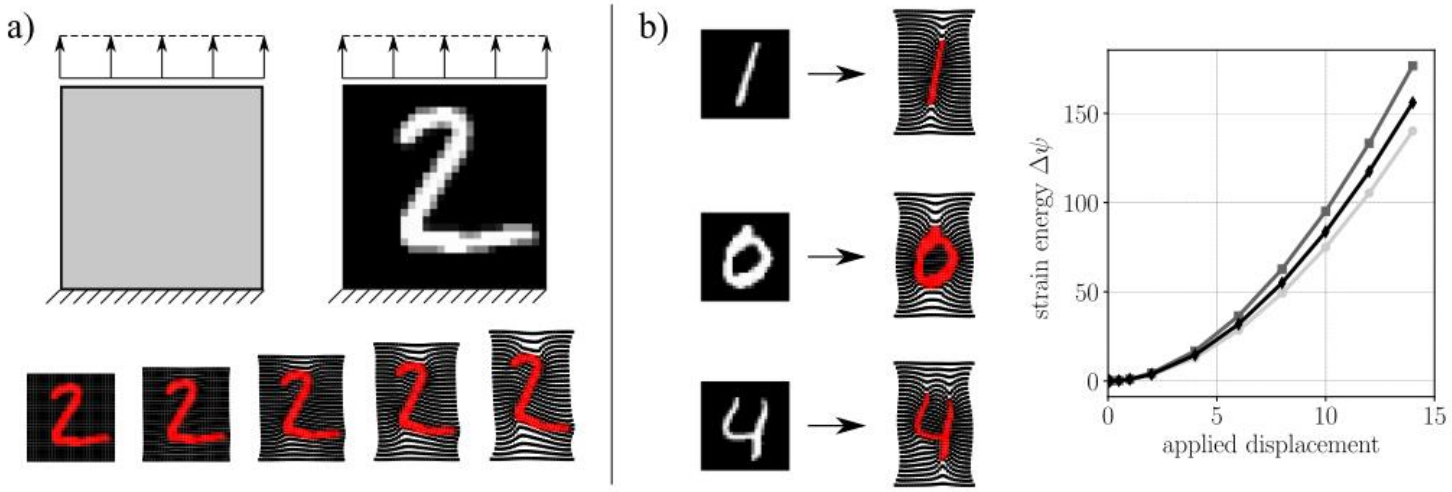


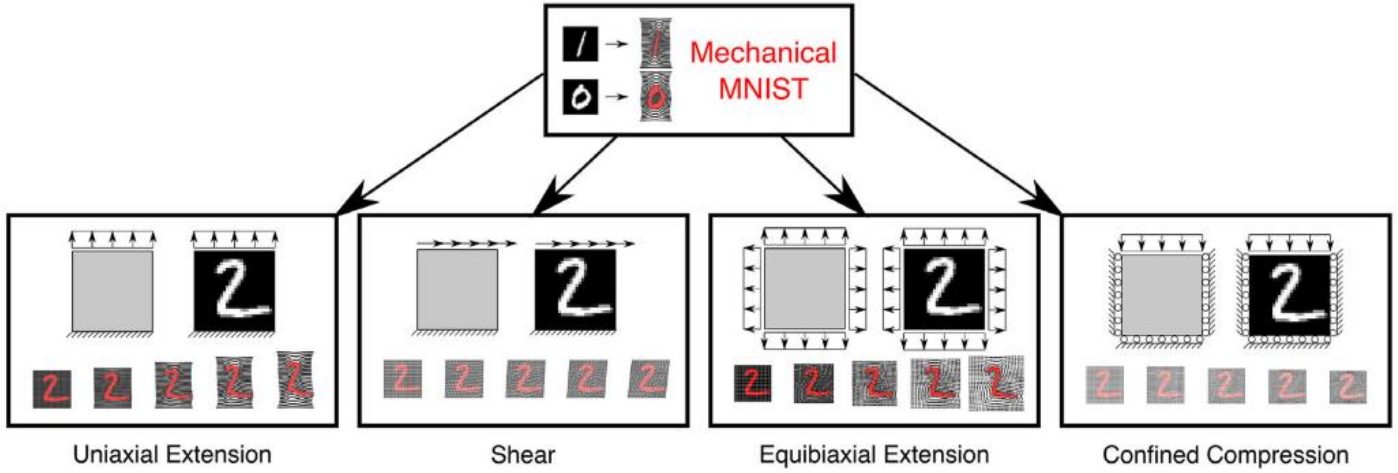
➢ Examples



Liu, Z. and Wu, C.T., "Exploring the 3D architectures of deep material network in data-driven multiscale mechanics," *J. Mech. Phys. Solids, 2019*.

Mechanical MNIST: A benchmark dataset for mechanical metamodels





- **Varied stiffness with grayscale value**

$$E = \frac{b}{255.0}(100.0 - 1.0) + 1.0$$

- **Various mechanical loading cases**
  - Uniaxial tension
  - Shear
  - Biaxial tension
  - Confined compression
- **Outputs**
  - Global displacement, force, energy
  - Full-filed displacement

Lejeune, E., "Mechanical MNIST: A benchmark dataset for mechanical metamodels," *Extrem. Mech. Lett.*, 2020

Summary

- Data-driven applications in mechanics
  - Constitutive model-free approach
  - Full-filed material identification
  - Hierarchical and mechanistic data-driven model
  - Multi-scale modeling